

Brilliant Compendium

Table 1: Embeddings

Method	Formula
Position Encoding	
Sinusoidal Position Encoding	$\text{PE}_{(\text{pos}, 2i)} = \sin\left(\frac{\text{pos}}{10000^{2i/d_{\text{model}}}}\right)$ $\text{PE}_{(\text{pos}, 2i+1)} = \cos\left(\frac{\text{pos}}{10000^{2i/d_{\text{model}}}}\right)$
Time Encoding	
Logit-Normal Distribution	$f(t; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \frac{1}{t(1-t)} \exp\left(\frac{(-\text{logit}(t) - \mu)^2}{2\sigma^2}\right)$ where, $\text{logit}(t) = \log\left(\frac{t}{1-t}\right)$
Fourier Time Encoder	$\text{FTE} : [0, 1] \rightarrow \mathbb{R}^{2d},$ $t \mapsto [\cos(2\pi w_1 t) \ \cdots \ \cos(2\pi w_d t) \ \sin(2\pi w_1 t) \ \cdots \ \sin(2\pi w_d t)]^T$ where, $w_i \sim \mathcal{N}(0, 1)$ are learnable.

Table 2: Divergences

Topic	Formula
Bregman Divergence	$D_f(x, y) = f(x) - (f(y) + \langle \nabla f(y), x - y \rangle)$